PowerPoint Fusion Assistant

Multimodal Interaction - 4th Assignment

Mariana Perna, 108067
Sara Almeida, 108796

Introduction and Project Objective

In this project, we developed an assistant to control PowerPoint presentations using a combination of voice and gestures, either simultaneously or independently, aiming to provide a more intuitive and seamless experience for presenters. The system uses a Kinect sensor to capture the presenter's movements, which are processed through a Generic Gesture Modality and sent to the Fusion Engine. The Fusion Engine handles the integration of gestures and voice commands using an SCXML-based configuration and communicates with the Interaction Manager (IM-MMI) to identify the requested actions. These actions are then sent to a C# backend, which interacts with the PowerPoint Interop API to execute commands like advancing slides, zooming in, or starting a timer. Additionally, voice commands are processed using Rasa, while the system provides immediate feedback through a Web Application, ensuring the presenter is always informed about the actions being performed. This multimodal approach simplifies the control of presentations, making the process more dynamic and accessible.

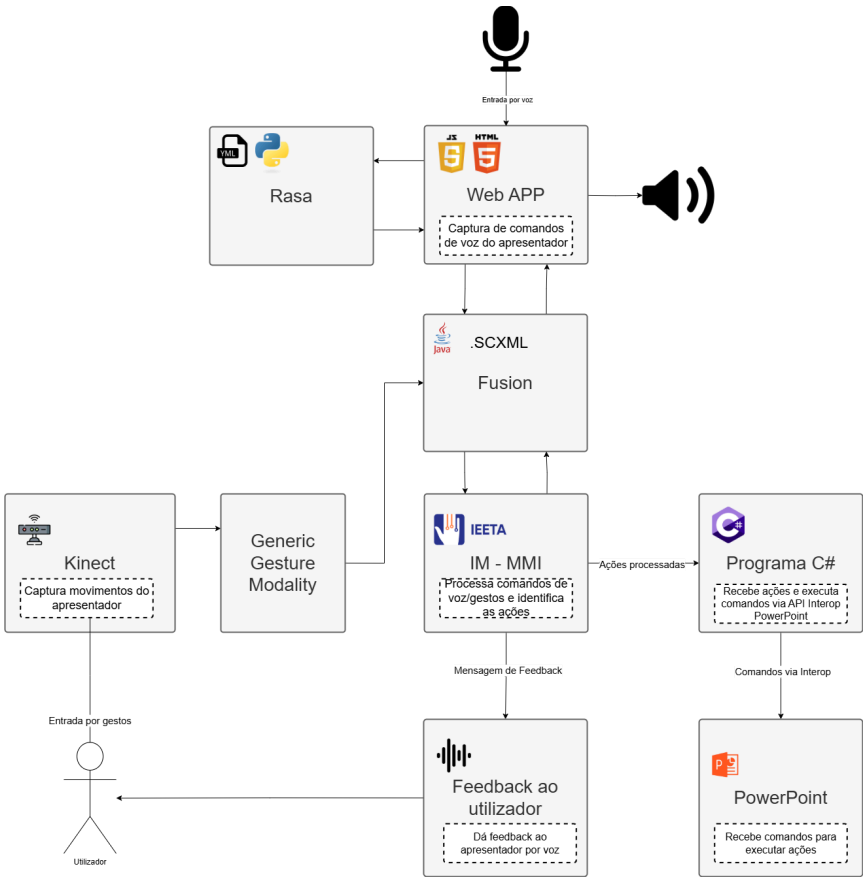


Figure 1: System Architecture

This assistant enhances the flow of presentations by enabling the presenter to perform various actions, such as starting, navigating, and ending a slideshow, using a combination of gestures and voice commands. This solution eliminates the need for manual interaction with the computer, offering a hands-free and dynamic approach to presentation control. It is particularly beneficial for presenters in scenarios requiring fluid interaction or for individuals with accessibility needs.

The primary objective of this project was to develop an intuitive and user-friendly system that leverages multimodal interaction to simplify the presentation process. By combining gestures and voice commands, the assistant ensures a seamless, efficient, and professional presentation experience, minimizing disruptions and maximizing presenter focus.

The following section details the integration of gestures and voice commands, their respective functionalities, and the system's responses. Each interaction was designed to be intuitive, providing immediate feedback.

Implemented Functionalities and System Feedback

Complementarity

Complementarity ensures that different modalities, in this case gestures and voice commands, work only together and with the goal of providing a seamless and effective interaction. This approach was particularly helpful for intents like checking elapsed time or requesting assistance, where the combined use of modalities makes the interaction intuitive and practical.

1. Show Elapsed Time

The 'show elapsed time' intent provides presenters with the total time elapsed since the beginning of the presentation, helping them manage their schedule and stay on track. We chose this feature to be a complementary interaction because sometimes, during a presentation or even when training them, the user may need to check the hours or maybe see a message they received in their smartwatch without wanting to be interrupted by the system feedback saying the time that has passed. This way, with the need for this gesture with the voice command that asks for time or something similar, we make sure that the user really wants to use this feature and know the elapsed time since the start of the presentation.

Example Commands:
"Quanto tempo já passou?"
"Qual o tempo desde o início?"
"Espero não estar a ultrapassar o tempo decorrido."

Gesture:
Raising the left arm as if to check the hours and pointing to the wrist with the right hand. (as if checking a watch)

System Feedback:
"Passaram: {elapsedTime.Hours} horas, {elapsedTime.Minutes} minutos e {elapsedTime.Seconds} segundos." -> *Voice Feedback*

2. Helper

The *'helper'* intent is designed to assist users, especially those new to the system, by providing guidance on how to use the available commands. This feature offers a simple way to learn the system's functionality by opening another explanatory PowerPoint when the user makes the gesture and asks for help using voice. We came up with this solution thinking it would be ideal for first-time users or anyone in need of a quick refresher. The user can then return to their actual PowerPoint at the point they left it by using the voice command *end_helpe*r, explained later in this report.

Example Commands:
"Preciso de ajuda"
"Que comandos existem?"

Gesture:
Right hand on the face and left hand on the elbow.

System Feedback:
"Olá! Estou aqui para ajudar! Aqui está um powerpoint onde podes ver tudo o que podes fazer para interagir com o sistema!" -> *Voice Feedback*

Redundancy

Redundancy in our system refers to the ability to perform the same action using different modalities, ensuring flexibility and reliability during presentations. This approach guarantees that users can seamlessly navigate their slides, even if one modality is unavailable or less convenient in a given situation.

1. Next Slide

The '*next slide*' intent is a core functionality of the system, allowing presenters to advance their slides effortlessly. This feature eliminates the need for physical interaction with the computer, ensuring a smoother presentation flow and enabling the presenter to remain engaged with the audience. By providing both gesture and voice command options, the system leverages redundancy to make this action reliable and accessible in different scenarios.

Example Commands:
"Próximo slide, por favor"
"Avançando para o próximo slide"
"Já terminei este"
"Estou farta deste slide passa para o próximo"

Gesture:
Raise the forearm and point your thumb to the left.

System Feedback:
The slideshow shows the next slide.
"Ok!" -> *Voice Feedback*

2. Previous Slide

The *'previous slide'* intent is essential for revisiting important points or addressing audience questions. Similar to the 'next slide' feature, it removes the need for physical interaction with the computer, helping the presenter maintain eye contact and focus on their delivery. Offering both gesture and voice command options ensures flexibility and reinforces the seamless flow of the presentation.

Example Commands:
"Slide anterior"
"Esqueci-me de dizer uma coisa no slide anterior"
"Vai para trás"

Example Commands:
Raise the forearm and point your thumb to the right.

System Feedback:
The slideshow shows the previous slide.
“Feito!” -> *Voice Feedback*

Single-Modality Interactions

- Voice commands:

1. Jump to Slide by Title

The *‘jump to slide by title’* intent allows users to directly navigate to a specific slide by providing its title. This feature is particularly useful in large or complex presentations or during the Q&A sessions when a particular topic needs to be revisited.

Example Commands:
"Passando aos [Resultados]."
"Voltando ao slide da [Introdução]."
"Mostra-me o slide da [Introdução], por favor"

System Feedback:
"Slide de {title}"
"Slide com o título {title} não encontrado."

2. Jump to Slide by Number

The *‘jump to slide by number’* intent was chosen to provide an alternative navigation method for users who prefer or remember slide numbers instead of titles.
This feature enhances the system’s versatility, catering to different user preferences.

Example Commands:
"Passando para o slide número [3]."
"Mostra-me o slide número [5], por favor."

System Feedback:
"Slide {slideNumber}."
"Slide número {slideNumber} não encontrado.”

3. Highlight Phrase

The *'highlight phrase'* intent was designed to enable dynamic emphasis on key points during a presentation by highlighting text. This functionality is particularly valuable for drawing the audience's attention to specific content, enhancing visual impact.

Example Commands:
"É de destacar que [phrase]"
"Sublinhando que [phrase]"

System Feedback:
"Texto '{phraseToHighlight}' destacado."
"Não foi possível destacar o texto '{phraseToHighlight}'."

4. Zoom In

The *'zoom in'* intent allows users to focus on specific areas of a slide, such as an image or graph, by enlarging them for better visibility. This functionality improves audience engagement by focusing attention on key visual elements.

Example Commands:
"Dá zoom na imagem."
"Ampliando a imagem."
"Quero ver melhor esta imagem."

System Feedback:
"Zoom aplicado."
"Nenhuma área principal foi encontrada no slide para aplicar zoom."

5. Zoom Out

The *'zoom out'* intent complements the previous one allowing the users to revert to the original slide view after using the *'zoom in'* functionality.

Example Commands:
"Reduz o zoom."
"Voltando ao tamanho original."

System Feedback:
"Zoom revertido."

6. Current Slide

The *'current slide'* intent provides the user with information about their current position in the presentation, helping them to stay oriented, especially in larger slide decks where it can be easy to lose track of progress.

Example Commands:
"Em que slide estou?"
"Qual é o slide atual?"
"Onde estou na apresentação?"

System Feedback:
"Estás no slide número 5."

7. Slides Left

The *'slides left'* intent was included to provide users with an immediate overview of their progress in the presentation. This is particularly useful for time management, as it helps presenters determine how much content remains to be covered.

Example Commands:
"Quantos slides faltam?"
"Quanto falta para terminar a apresentação?"

System Feedback:
""Ainda faltam {slidesLeft} slides para terminar a apresentação."
"Você está no último slide."

8. Restart Presentation

The *'restart presentation'* intent allows users to restart their presentations from the first slide and it was implemented with the thought that it would be useful when users are training their presentation or after an interruption. It is particularly helpful in academic or professional presentations where timing and structure are being practiced.

Example Commands:
"Reinicia a apresentação"
"Começa do início"
"Volta ao início"

System Feedback:
"Reiniciada"

9. Start Timer & Stop Timer

The *'start timer'* and *'stop timer'* intents allow users to start and stop a timer, as the name reveals, and they were added to help presenters practice timing specific sections of their presentation. These features are particularly valuable during the training phase.

Example Commands:
"Inicia o temporizador"
"Para o temporizador"

System Feedback:
"Temporizador iniciado."
"Temporizador parado depois de {elapsed.Hours} horas, {elapsed.Minutes} minutos e {elapsed.Seconds} segundos."

10. Close Presentation

The *'close presentation'* intent gives users a convenient way to end their presentation through voice commands. This eliminates the need for manual interaction, maintaining the flow of the session.

Example Commands:
"Terminamos assim a apresentação."
"Termina a apresentação."

System Feedback:
"Apresentação fechada com sucesso."

11. End_helper

The *'end helper'* intent gives users a convenient way to end the help PowerPoint when they no longer need it. This way they can go back to their presentation where they left it.

Example Commands:
"Obrigada! Já não preciso."
"Obrigada! Já percebi."

System Feedback:
"De nada! Boa sorte!"-> *Voice Feedback*

- Gestures:

1. Start Presentation

The ‘start presentation’ gesture was implemented to allow presenters to initiate their slideshow seamlessly without requiring manual interaction. This feature ensures a smooth and professional start to the presentation, eliminating distractions and providing a hands-free solution.

Gesture:
Pointing up and a bit to the left, with the left arm and the hand open.

System Feedback:
The presentation and presenter view are initiated.

2. Finish Presentation

The ‘end presentation’ gesture was implemented to allow presenters to conclude their slideshow in a natural and elegant manner, without needing to manually interact with the computer. This gesture adds a professional touch to the presentation experience.

Gesture:
Bowing with the body.

System Feedback:
The presentation is finished and closed.

3. Advance Two Slides

The ‘advance two slides’ gesture was chosen to enable quick navigation during presentations, offering an efficient way to skip over less relevant slides without manual interaction, in case the presenter is running out of time, for example. This feature enhances the system’s usability, catering to scenarios where the presenter needs to adjust the flow dynamically and maintain audience engagement.

Gesture:
Gesture of "two fingers" raised at the belly level, with the left hand.

System Feedback:
Success: Advances 2 slides.
Error (e.g., insufficient slides remaining): "Não é possível saltar 2 slides. Já está no final da apresentação." -> *Voice Feedback*

4. Request Silence

The ‘request silence’ gesture allows the presenter to politely ask the audience for silence in a non-disruptive and natural way. This feature is particularly useful in situations where the presentation includes more than one presenter and we don’t want to interrupt our colleague talking and so this is crucial to maintain focus and reducing distractions.

Gesture:
"Shhh" gesture (hand on mouth with the right hand index finger raised).

System Feedback:
“Pedimos silêncio à audiência, por favor!” -> *Voice Feedback*

5. Does anyone have questions

The ‘does anyone have questions’ gesture enables the presenter to smoothly transition into a Q&A session, engaging the audience and encouraging interaction. This gesture provides a natural and intuitive way to open the floor for questions without needing manual intervention.

Gesture:
Open your arms outstretched forward.

System Feedback:
“Alguém tem alguma dúvida?” -> *Voice Feedback*

Conclusion

The integration of gestures and voice commands in our system highlights its ability to provide a seamless, intuitive, and efficient solution for controlling PowerPoint presentations. We implemented **11** voice intents; **5** gestures interaction; and **4** multimodal interactions with fusion. With these combinations of interactions, the system offers flexibility and accessibility, catering to diverse presentation needs while reducing physical interaction with the computer. Each functionality was carefully designed, implemented, and tested to ensure reliability and responsiveness, enhancing the presenter’s experience. This project demonstrates how multimodal interaction can simplify tasks and create not only more engaging but also more accessible presentation experiences. The knowledge gained from this development sets a strong foundation for future improvements, where we aim to further optimize and expand the system’s capabilities with Fusion.